

# External Fixture PatchTST Harm Diagnostic

Adaptive router on external forecasts; MSE delta is percentage vs. no correction. Positive values indicate harm.

| Dataset       | Horizon   | Router MSE Δ (%) | Matched smoothing Δ (%) | Correction rate (%) | Smoothing action (%) | Harmed |
|---------------|-----------|------------------|-------------------------|---------------------|----------------------|--------|
| ETTh1         | 96        | 0.411            | 0.025                   | 56.2                | 25.0                 | Yes    |
| ETTh1         | 192       | 0.105            | -0.027                  | 75.0                | 46.9                 | Yes    |
| ETTh1         | 336       | 0.088            | -0.024                  | 78.1                | 50.0                 | Yes    |
| ETTh1         | 720       | -0.217           | -0.021                  | 75.0                | 50.0                 | No     |
| ETTm1         | 96        | -0.465           | -0.354                  | 21.9                | 12.5                 | No     |
| ETTm1         | 192       | -0.007           | -0.126                  | 18.8                | 6.2                  | No     |
| ETTm1         | 336       | -0.024           | -0.250                  | 15.6                | 6.2                  | No     |
| ETTm1         | 720       | 0.144            | -0.037                  | 12.5                | 9.4                  | Yes    |
| PatchTST avg. | 8 configs | 0.004            | -0.105                  | 44.9                | 25.8                 | 4/8    |
| DLinear avg.  | 8 configs | -1.865           | -0.391                  | 44.5                | 22.7                 | 0/8    |

Setup: external fixture = 2 datasets (ETTh1, ETTm1) x 2 backbones (DLinear, PatchTST) x 4 horizons (96, 192, 336, 720), 32 samples per config. Matched smoothing uses the same correction rate as the router. Interpretation: PatchTST shows weak average improvement and 4/8 harmed configs, while DLinear improves on average with 0/8 harmed configs.